The Human Heart and Circulatory System

Introduction to the Circulatory System

The circulatory system, also called the cardiovascular system, is the body's transport network. It consists of the heart, the blood vessels and the blood. Its job is to carry oxygen and nutrients to every cell, remove carbon dioxide and other waste products, deliver hormones, help fight infection and help regulate body temperature. An adult body contains about 5 litres of blood, and the heart pumps all of it around the body in about one minute. The heart beats about 60 to 100 times per minute at rest, roughly 100,000 times a day and more than 2.5 billion times in an average lifetime. The English physician William Harvey described the complete circulation of blood in 1628, showing that the heart pumps blood in a continuous circuit.

The Human Heart

The heart is a muscular organ, about the size of a person's fist, weighing roughly 250 to 350 grams, located in the chest slightly to the left of the centre, between the lungs and protected by the rib cage. Its wall is made of a special muscle called cardiac muscle that contracts rhythmically without tiring, and it is surrounded by a protective sac, the pericardium. The heart's own muscle receives blood through the coronary arteries.

How Many Chambers Does the Heart Have?

The human heart has four chambers. The two upper chambers are the right atrium and the left atrium, which receive blood coming into the heart. The two lower chambers are the right ventricle and the left ventricle, which pump blood out of the heart. The left ventricle has the thickest muscular wall because it pumps blood to the whole body, whereas the right ventricle pumps blood only to the lungs. A wall called the septum separates the right and left sides so that oxygen-rich and oxygen-poor blood do not mix. Four valves keep blood flowing in one direction: the tricuspid valve between the right atrium and right ventricle, the pulmonary valve at the exit to the lungs, the mitral (bicuspid) valve between the left atrium and left ventricle, and the aortic valve at the exit to the aorta. The "lub-dub" sound of the heartbeat is made by the closing of these valves. Fish have a two-chambered heart, amphibians and most reptiles three, while birds and mammals have four.

Double Circulation

Humans have double circulation. In pulmonary circulation, oxygen-poor blood from the body enters the right atrium, goes to the right ventricle and is pumped through the pulmonary artery to the lungs, where it picks up oxygen and gives up carbon dioxide. In systemic circulation, the oxygen-rich blood returns through the pulmonary veins to the left atrium, moves to the left ventricle and is pumped through the aorta, the largest artery, to the rest of the body. Note that the pulmonary artery is the only artery carrying deoxygenated blood, and the pulmonary vein is the only vein carrying oxygenated blood.

Blood Vessels

Arteries carry blood away from the heart; they have thick, elastic walls to withstand high pressure. Veins carry blood back to the heart; they have thinner walls and valves to prevent backflow. Capillaries are tiny, thin-walled vessels where the exchange of oxygen, nutrients and wastes between blood and tissues takes place. The total length of all blood vessels in the body is about 100,000 kilometres.

Blood and What Red Blood Cells Carry

Blood has four components: plasma, red blood cells, white blood cells and platelets. Plasma is the straw-coloured liquid, about 55 percent of blood, which carries water, salts, proteins, hormones and nutrients. Red blood cells (RBCs), also called erythrocytes, are disc-shaped cells without a nucleus, made in the bone marrow, which live about 120 days. Red blood cells carry oxygen from the lungs to the tissues, using a protein called haemoglobin, which contains iron and gives blood its red colour; they also carry some carbon dioxide back to the lungs. A shortage of haemoglobin or red cells is called anaemia, which is common in India and causes tiredness. White blood cells defend the body against infections. Platelets help the blood to clot and stop bleeding. Blood groups A, B, AB and O were discovered by Karl Landsteiner in 1901.

What Is Normal Blood Pressure?

Blood pressure is the force of blood against the walls of the arteries, measured in millimetres of mercury (mmHg) using a sphygmomanometer. It is written as two numbers: systolic pressure, when the heart contracts, over diastolic pressure, when the heart relaxes. Normal blood pressure for a healthy adult is around 120/80 mmHg, with readings below 120 over below 80 considered optimal. High blood pressure, or hypertension, is generally diagnosed when readings are repeatedly 140/90 mmHg or higher (some guidelines use 130/80). Hypertension is called the silent killer because it often has no symptoms, yet it raises the risk of heart attack, stroke, kidney disease and heart failure. Low blood pressure, below about 90/60, can cause dizziness.

Heart Diseases and Prevention

Coronary artery disease occurs when the arteries supplying the heart are narrowed by fatty deposits (atherosclerosis), and a blockage causes a heart attack. Other conditions include hypertension, heart failure, arrhythmias and congenital defects. Risk factors include smoking, high cholesterol, diabetes, obesity, a sedentary lifestyle, stress and family history. Cardiovascular diseases are the leading cause of death in India and the world. Prevention includes a balanced diet low in salt, sugar and trans fat, at least 150 minutes of moderate exercise a week, no tobacco, controlling weight, regular check-ups and managing stress. Warning signs of a heart attack include chest pain or pressure, pain spreading to the arm or jaw, sweating, breathlessness and nausea; immediate emergency care is vital. World Heart Day is observed on 29 September.

The Heart in India

India has pioneered low-cost heart care, and Chennai is a major centre of cardiac surgery and a medical tourism hub. India's first successful human heart transplant was performed at AIIMS, New Delhi, in August 1994 by Dr. P. Venugopal. Indian companies make affordable heart stents, and the government regulates their prices. Indian doctors popularise screening for hypertension and diabetes in primary health centres.

Challenges and Future

Challenges include rising lifestyle diseases and low awareness. The future includes wearable ECG devices in smartwatches, artificial hearts, AI-based diagnosis from ECGs and gene therapy.

Quick Facts

- The human heart has four chambers: two atria and two ventricles.
- Normal blood pressure is about 120/80 mmHg.
- Red blood cells carry oxygen using haemoglobin.
- William Harvey described blood circulation in 1628.
- World Heart Day is celebrated on 29 September.